# Final Exam (Solution)

**Intro to Probability and Statistics**
**EE 354 / CE 361 / MATH 310**

Fall 2021

**Question 1** (10 points)

(a)

Suppose that a point $(x, y)$ is chosen from the square S = [-1,1] x [-1,1] with all the points in S being equally likely to be chosen.

i. What is the probability that the point (0,0) is chosen?
ii. What is the probability that the point chosen falls in the square $S_2$ = [0,1] x [0,1]?

(b)

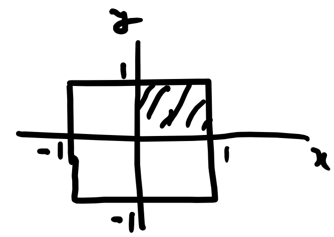
A Habib University student has 5 books about Mathematics, 4 books about Computer Science, and 2 books about Economics. The student decides to randomly arrange the books on a bookshelf. What is the probability that the randomly-created book arrangement ends up containing all books from a given subject area contiguous to each other?

**Solution:**

**(a)**

i)

$$P\{(0,0) \text{ is chosen}\} = 0$$




ii)

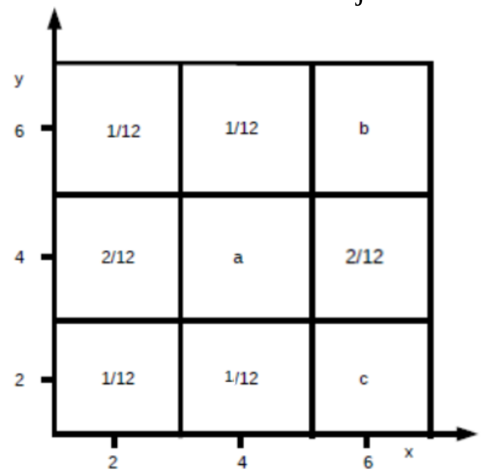
$$P\{\text{Chosen point is in the square } [0,1]\times[0,1]\} = \frac{1}{4}$$

(b)

$$\text{Required Probability} = \frac{3!\,5!\,4!\,2!}{11!}$$

**Question 2** (10 points)

Let X and Y be two discrete random variables with their joint PMF shown below:




a) Calculate $p_X(2)$.
b) Find and sketch $p_{Y|X}(y|2)$
c) Calculate $E[Y|X=2]$
d) Determine the values of a, b, and c, under which X and Y are independent random variables. Justify your answer.

(a)

$$p_X(x) = \sum_y p_{X,Y}(x,y)$$

$$\Rightarrow p_X(2) = p_{X,Y}(2,2) + p_{X,Y}(2,4) + p_{X,Y}(2,6)$$

$$= \frac{1}{12} + \frac{2}{12} + \frac{1}{12}$$

$$p_X(2) = \frac{1}{3} \quad \text{— Ans}$$

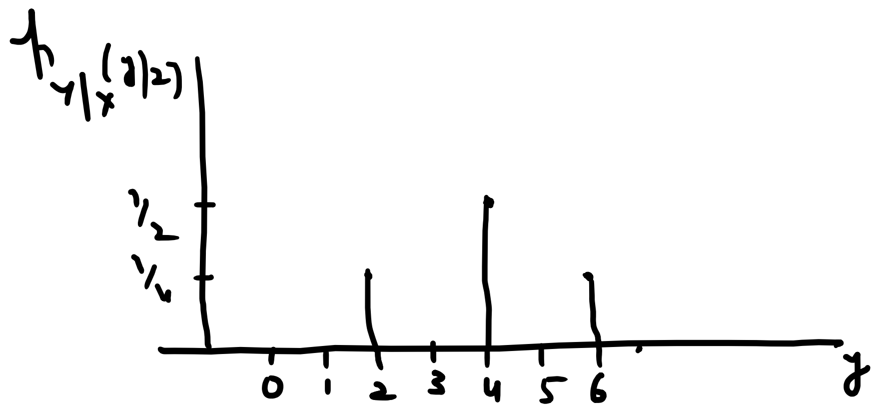
(b)

$$p_{Y|X}(y|x) = \frac{p_{X,Y}(x,y)}{p_X(x)}$$

$$p_{Y|X}(2|2) = \frac{p_{X,Y}(2,2)}{p_X(2)} = \frac{1/12}{1/3} = \frac{1}{4}$$

$$p_{Y|X}(4|2) = \frac{p_{X,Y}(2,4)}{p_X(2)} = \frac{2/12}{1/3} = \frac{1}{2}$$

$$p_{Y|X}(6|2) = \frac{p_{X,Y}(2,6)}{p_X(2)} = \frac{1/12}{1/3} = \frac{1}{4}$$




(c)

$$E[Y|X=2] = 2\left(\frac{1}{4}\right) + 4\left(\frac{1}{2}\right) + 6\left(\frac{1}{4}\right)$$

$$= \frac{1}{2} + 2 + \frac{3}{2}$$

$$E[Y|X=2] = 4$$ — Ans

(d)

$a = \frac{2}{12}$ $b = \frac{1}{12}$ $c = \frac{1}{12}$

With these choices, all the rows have the same relative likelihood among the members of the row. This implies that having information about y does not change the relative likelihoods of x's. This implies independence.

Similar argument can be applied about columns as well.

**Question 3** (10 points)

(a)

If the height of Habib University students is modeled as a Normal random variable with mean of 65 inches and standard deviation of 3 inches, what is the probability that a randomly chosen HU student has a height of 6ft or more?

(b)

X and Y are random variables with a joint PDF given by:

$$f_{X,Y}(x,y) = \frac{2}{c}$$

for $0 \le x \le 1$ and $0 \le y \le 1$. The joint PDF is zero for all other values of x and y.

i. Find the value of c.
ii. Calculate the marginal PDFs of X and Y.

(a)

Let H be the random variable representing the height of HU students.

$$H \sim N(65, 9)$$

$$P(H \geq 72) = P\left(\frac{H-65}{3} \geq \frac{72-65}{3}\right)$$

$$= P(Y \geq 2.33)$$ where $Y \sim N(0,1)$

$$= 1 - P(Y \leq 2.33)$$

$$= 1 - 0.9901$$

$$P(H \geq 72) = 0.0099$$ — Ans

(b)

i)

$$f_{X,Y}(x,y) = \begin{cases} \frac{2}{c} & 0 \le x \le 1,\ 0 \le y \le 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\because \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{XY}(x,y)\, dx\, dy = 1$$

$$\Rightarrow \quad f_{X,Y}(x,y) = \begin{cases} 1 & 0 \le x \le 1,\ 0 \le y \le 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\Rightarrow \quad c = 2$$

ii)

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\, .dy$$

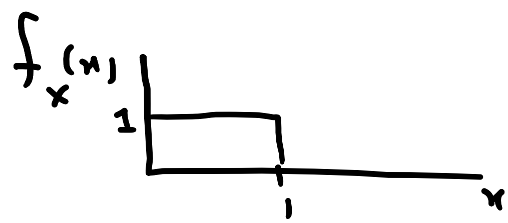
$$= \int_0^1 1\, .dy = y\Big|_0^1$$

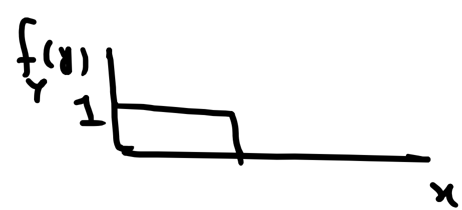
$$f_X(x) = 1 \qquad 0 \le x \le 1$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\, dx$$

$$= \int_0^1 1\, .dx = x\Big|_0^1$$

$$f_Y(y) = 1 \qquad 0 \le y \le 1$$



**Question 4** (10 points)

W and X are independent continuous random variables that are uniformly distributed with W$\sim U[0,1]$ and X$\sim U[0,2]$. Find the Cumulative Distribution Function, CDF, and Probability Density Function, PDF, for the following random variables:

a) $Y = \max(W, X)$

b) $Z = \max(W, X) + 1$

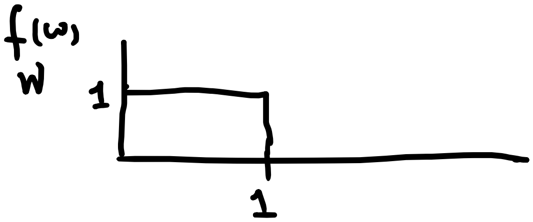
(a)

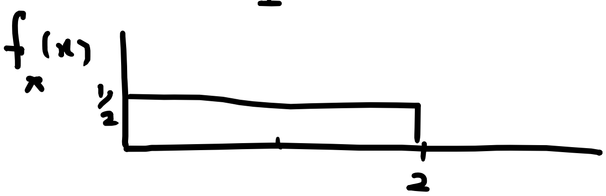
$$F_Y(y) = P(Y \le y) = P(\max(W,X) \le y)$$

$$= P(W \le y,\ X \le y)$$

$\because$ W & X are independent

$$F_Y(y) = P(W \le y) \cdot P(X \le y) \quad \text{—(1)}$$




Now

$$P(W \le y) = \begin{cases} 0 & y < 0 \\ y & 0 \le y < 1 \\ 1 & 1 \le y \end{cases}$$

$$P(X \le y) = \begin{cases} 0 & y < 0 \\ y/2 & 0 \le y < 2 \\ 1 & 2 \le y \end{cases}$$

From (1)

$$F_Y(y) = \begin{cases} 0 & y < 0 \\ y^2/2 & 0 \le y < 1 \\ y/2 & 1 \le y < 2 \\ 1 & 2 \le y \end{cases}$$

$\because \quad f_Y(y) = \frac{d}{dy}\left[F_Y(y)\right]$

$$f_Y(y) = \begin{cases} 0 & y < 0 \\ y & 0 \le y < 1 \\ 1/2 & 1 \le y < 2 \\ 0 & 2 \le y \end{cases}$$

(b)

Note that

$$Z = Y + 1$$

$\Rightarrow$ $F_Z(z)$ and $f_Z(z)$ are just shifted (by 1) versions of $F_Y(y)$ and $f_Y(y)$

**Question 5** (10 points)

On a sunny day, it takes a student between 30 to 45 minutes to get to Habib University campus with all times being equally likely. On a rainy day, it takes the same student between 40 to 60 minutes to get to campus with all times being equally likely. Assume a day is sunny with probability 0.75 and rainy with probability 0.25.

a) Find the PDF of the time T that it takes the student to get to campus.

b) On a given day, it took the student 42 minutes to get to campus. What is the probability that this particular day was rainy?

(a)

Let X be the random variable that takes the value 0 when its a sunny day and value 1 when its a rainy day.

$$p_X(x) = \begin{cases} 0.75 & \text{when } x = 0 \\ 0.25 & \text{when } x = 1 \end{cases}$$

Now,

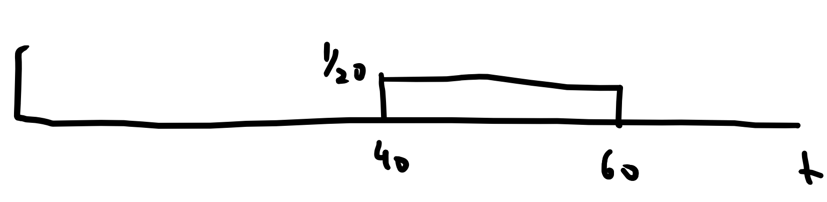
$$f_T(t) = \sum_x p_X(x) f_{T|X}(t|x)$$

$$= p_X(0) f_{T|X}(t|0) + p_X(1) f_{T|X}(t|1)$$

$$f_T(t) = 0.75 \left[ U(30,45) \right] + 0.25 \left[ U(40,60) \right] \quad \text{— (1)}$$

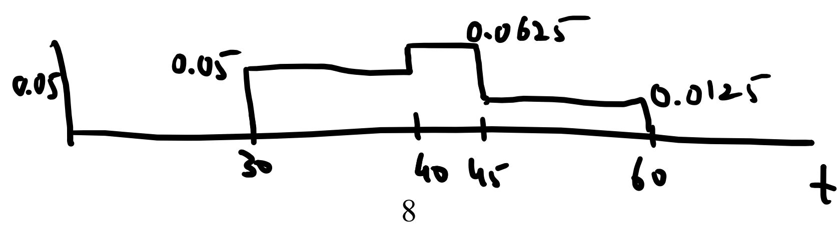
$f_{T|X}(t|0)$


$f_{T|X}(t|1)$




From 1,

$f_T(t)$



(b)

$p_{X|T}(1|42) = ?$

Using Bayes' Rule :- (Discrete X, Continuous Y)

$$p_{X|Y}(x|y) = \frac{p_X(x)\, f_{Y|X}(y|x)}{f_Y(y)}$$

$$\Rightarrow \quad p_{X|T}(x|t) = \frac{p_X(x)\, f_{T|X}(t|x)}{f_T(t)}$$

$$p_{X|T}(1|42) = \frac{p_X(1)\, f_{T|X}(42|1)}{f_T(42)} \quad \text{--- (2)}$$

$p_X(1) = 0.25$

$f_{T|X}(42|1) = \frac{1}{20} = 0.05$

$f_T(42) = 0.0625$

From (2)

$$p_{X|T}(1|42) = \frac{0.25 \times 0.05}{0.0625}$$

$p_{X|T}(1|42) = 0.2$ —— Ans